Service Publication

Diesel Engine
8V 396 TE54

Description and
Operation Manual
M011055/00E

DETROIT DIESEL

mtu

DaimlerChrysler Powersystems Off-Highway

Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- Table of Contents (general)
- Table of Contents (section)
- Group / Subgroup
- Heading

Using the Alphabetical Index to trace an entry

- Term in Alphabetical Index
- Group / Subgroup
- Heading

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M011055/00E
Description and Operation Manual
8V 396 TE 54

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GERMANY

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Contents

Diesel Engine

Part A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUT, ENGINE POWER

Key to Engine Model Designation 8 V 396 TE 54

8 = number of cylinders

V = Vee configuration

396 = engine series - 100 times displacement of one cylinder in liters

T = exhaust turbocharging

E = charge air cooling in split-circuit coolant system

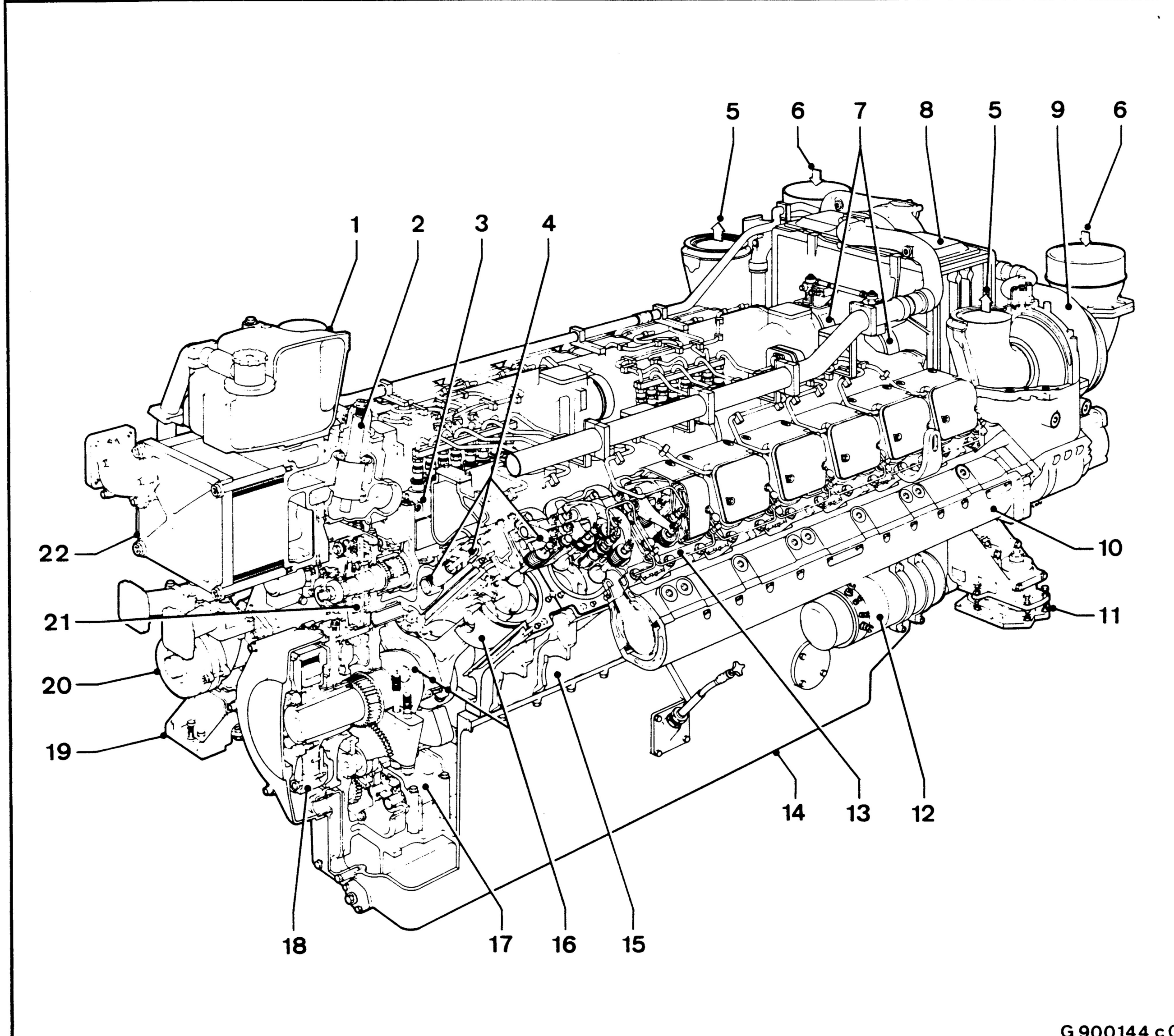
5 = marine engine

4 = design digit

Engine Power

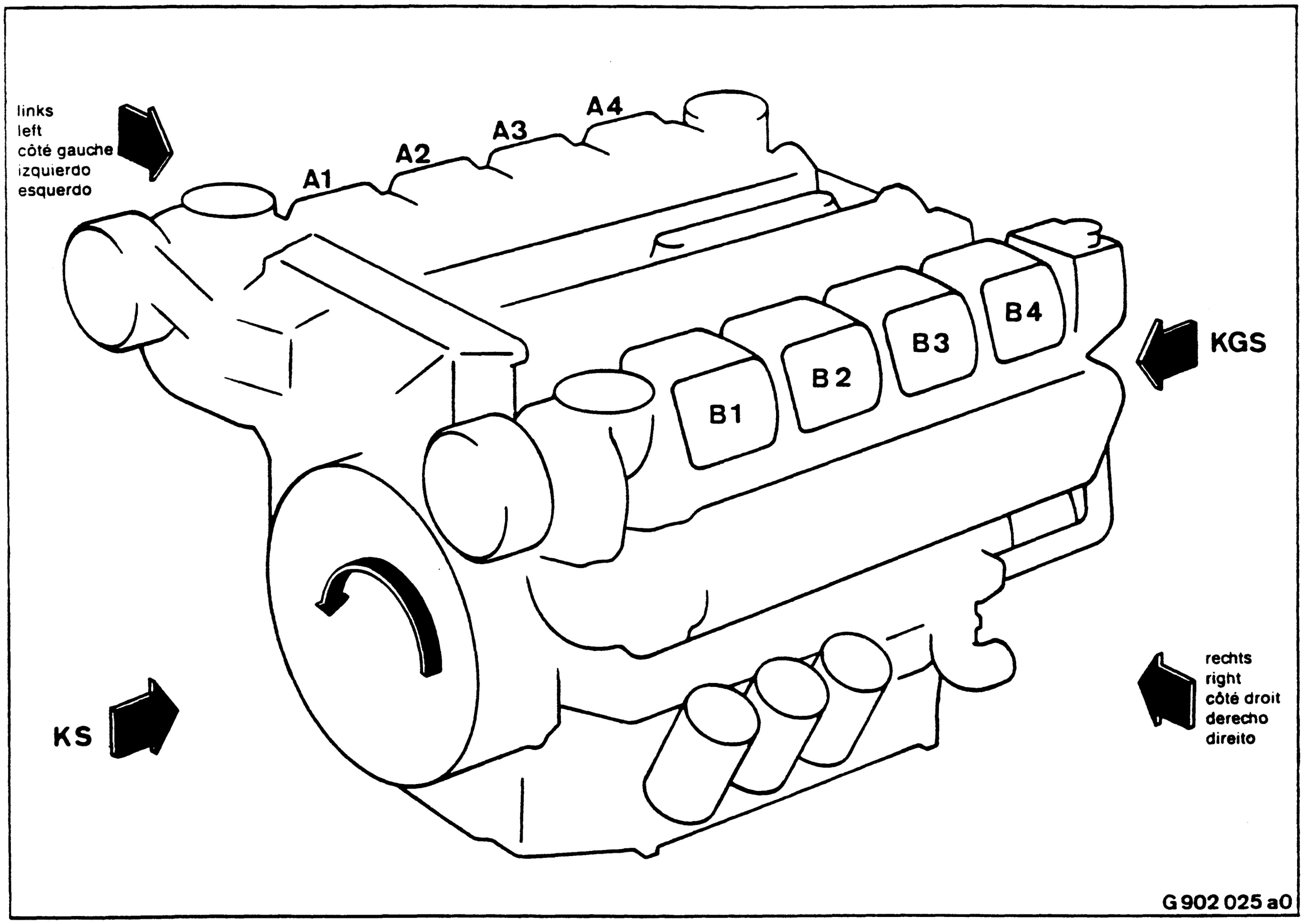
Continuous power as per ISO 3046	870 kW at 1800 rpm
- Overload capability for transient load response	10%
Reference conditions	
- intake air temperature	45°C
- raw water temperature	32°C
- barometric pressure	1000 mbar

Engine Layout Cutaway Drawing 16V 396 TE.4 (also applies to 8 and 12V)



1 Coolant expansion tank
2 Coolant thermostat
3 Fuel injection pump
4 Valve gear
5 Exhaust outlet
6 Air inlet
7 Emergency air shut-off flap
8 Intercooler
9 Exhaust turbocharger
10 Exhaust manifold
11 Engine mount, flywheel end
12 Starter
13 Cylinder head
14 Oil pan
15 Crankcase
16 Running gear
17 Engine oil pump
18 Vibration damper
19 Engine mount, timing end
20 Bilge pump
21 Fuel injection timer
22 Coolant cooler

ENGINE SIDE AND CYLINDER DESIGNATIONS



Engine sides are always designated as viewed from the flywheel end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN standard 6265).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the flywheel end.

Numbering of other engine components is also from the flywheel end, starting with No. 1.

Designations and abbreviations

Flywheel end = KS (formerly HKS)

Timing end = KGS (formerly GKS)

Left side (formerly A side)

Right side (formerly B side)

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	exhaust turbocharging
Cooling method	liquid cooling
Cylinder arrangement	90° Vee
Bore	165 mm
Stroke	185 mm
Displacement, cylinder	3.96 liters
Number of cylinders	8
Displacement, total	31.6 liters
Compression ratio	15.5 : 1
Direction of rotation (viewed from flywheel end)	c.c.w. (non-reversible)
Firing order	A1 A4-A2 A3 B4 B3 B2-B1
Fuel injection pressure:	
Setting pressure with new spring	310 to 318 bar
Specified pressure with run-in spring	300 to 308 bar
Minimum pressure with run-in spring	290 bar
Final compression pressure at 120 rpm, 1000 mbar and engine coolant at 5°C	24 to 28 bar
Break-away torque at engine coolant temperature of 40°C, allowance for acceleration included	approx. 400 Nm
Cranking torque at engine coolant temperature of 40°C	approx. 420 Nm
Firing speed at engine coolant temperature of 40°C	approx. 120 rpm
Mean piston speed at 1800 rpm	11.10 m/s
Engine noise without fan Mean air/free-field level, 1 m distance	104 dB (A)
Exhaust noise, unsilenced Air/free-field level, 1 m from outlet	118 dB (A)